Solving (1) and (2) we get

$$a = 50 \qquad b = 250$$

$$f(x) = \begin{cases} \frac{1}{b-a} = \frac{1}{250-50} = \frac{1}{200}; 50 < x < 250 \\ 0 \qquad\qquad otherwise \end{cases}$$

Example 10.21

Electric trains on a certain line run every 15 minutes between mid- night and six in the morning. What is the probability that a man entering the station at a random time during this period will have to wait at least ten minutes?

Let the random variable X denote the waiting time (in minutes).

The given assumption indicates that X is distributed Uniformly on (0,15).

$$P(X > 10) = \int_{10}^{15} f(x)dx = \frac{1}{15}\int_{10}^{15} 1dx$$

$$= \frac{1}{15}(x)_{10}^{15} = \frac{1}{15}(15-10) = \frac{1}{3}$$

10.2.2 NORMAL DISTRIBUTION

Introduction

The Normal distribution was first discovered by the English Mathematician De-Moivre in 1733, and he obtained this distribution as a limiting case of Binomial distribution and applied it to problems arising in the game of chance. In 1774 Laplace used it to estimate historical errors and in 1809 it was used by Gauss as the distribution of errors in Astronomy. Thus throughout the 18th and 19th centuries efforts were made for a common law for all continuous distributions which was then known as the Normal distribution.

Definition

A random variable X is said to have a Normal distribution with parameters μ and σ^2 if its probability density function is given by $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left\{\frac{x-\mu}{\sigma}\right\}^2}$ where

$-\infty < x < \infty$, $-\infty < \mu < \infty$ and $\sigma > 0$

It is denoted by $X \sim N(\mu, \sigma^2)$

Here μ is called as mean and σ^2 is variance of the distribution.